

PragmaJudge: a cross-disciplinary architecture for physical intelligence accountability

PragmaJudge can become the first system to cryptographically verify whether AI output faithfully served human intent — by treating the human body as an involuntary oracle. This report surfaces novel frameworks across eight disciplines that, when integrated, create an accountability architecture no deployed system currently provides. The convergence of three simultaneous shifts — AI agents that pass all existing software verification, regulatory mandates (EU AI Act Article 12) requiring tamper-evident AI audit trails, and biometric signals accurate enough to serve as cognitive ground truth — creates an unprecedented design window. PragmaJudge, built on IoTeX's DePIN infrastructure, sits at the exact intersection of these forces: physical intelligence verification, epistemic AI accountability, and decentralized mobile identity.

The research reveals a single architectural insight that unifies all eight domains: **the "illocutionary fidelity gap"** — the measurable distance between what a human intended (captured through involuntary physiological signals and speech-act-theoretic intent formalization) and what an AI delivered (evaluated by a cryptographically accountable agent fleet). Every framework below maps onto measuring, proving, or governing this gap.

The human body as an involuntary accountability oracle

The most architecturally significant finding across all domains is that **involuntary physiological signals can serve as ground-truth indicators of AI output quality** — and these signals cannot be faked by bots or AI agents. Three 2025 studies establish this empirically.

Kosmyna et al. (MIT Media Lab, 2025) found that humans passively accepting LLM output show **55% less neural connectivity** than those actively thinking — a measurable cognitive disengagement signature. (MIT Media Lab) (arXiv) Fišer et al. (PLOS ONE, 2025) demonstrated that AI-generated content triggers **higher GSR and wider pupil dilation** compared to human-created content, reflecting increased cognitive processing demands. (plos) A study in *Computers in Human Behavior Reports* (2025) showed that believing content is AI-generated triggers **decreased parasympathetic nervous system activity** — an automatic threat response beyond conscious control. (ScienceDirect)

These findings transform PragmaJudge's biometric signals from mere identity verification into cognitive accountability instruments. Galvanic skin response, previously treated as an emotional indicator, now functions as a **cognitive load sensor** (IEEE Xplore) with classification accuracy reaching **98.3%** using time-varying singular value decomposition (TSVD). (ScienceDirect) Keystroke dynamics (ScienceDirect) detect cognitive fatigue at **91% accuracy** and can distinguish deliberative typing (genuine evaluation) from uniform typing (rubber-stamping). (Fleksy) Mouse trajectory complexity correlates with decision conflict at the millisecond level — cursor path

analysis explains **68.8% of variance** in non-decision time and **32.82% of variance** in decision caution thresholds. (PubMed Central)

The critical insight for PragmaJudge: no single signal suffices, but **temporal cross-correlation across modalities** creates an unforgeable signature. When a human genuinely evaluates AI output, GSR onset timing, keystroke hesitation patterns, and mouse trajectory changes exhibit characteristic temporal binding. This multi-signal temporal pattern is essentially impossible for an automated system to replicate authentically, because it emerges from the involuntary coupling of sympathetic nervous system activation with motor planning circuits.

Cryptographic primitives that make intent verification possible

Four cryptographic constructions, none currently deployed together in any system, map directly onto PragmaJudge's Groth16/BN254 infrastructure and collectively solve the intent verification problem.

ZK-PoP (Zero-Knowledge Proof of Process) is the most directly applicable construction. Published in 2025, it generates Groth16 proofs over arithmetic circuits with Pedersen commitments to verify human authorship via behavioral biometrics. It proves three statements without revealing underlying data: sequential work function chains were computed correctly, behavioral feature vectors fall within human population distributions, and content evolution is consistent with incremental human editing. (ResearchGate) Performance is practical: **192-byte proofs** generated in under 30 seconds, verifiable in **8.2 milliseconds**, with less than 5% accuracy loss at privacy level $\epsilon \geq 1.0$. ZK-PoP already uses Groth16 — an exact match for PragmaJudge's proving system.

hinTS (threshold BLS signatures with silent setup), presented at IEEE S&P 2024, is ideally suited for PragmaJudge's 36-agent fleet. Each agent generates a BLS keypair independently; the joint public key derives deterministically with **no interactive distributed key generation required**. (IACR) (Hedera) A threshold of $t=24$ (matching the $0.65 \times 36 = 23.4$, rounded to 24) can be set dynamically post-setup. (IACR) Each agent signs its verdict in $\sim 1\text{ms}$. The aggregator collects ≥ 24 partial signatures and generates a SNARK proof of threshold satisfaction in ~ 0.5 seconds. (IACR) The final signature is **896 bytes** (Hedera) with constant **390K gas** verification cost on-chain — versus 60M gas for naive BLS multi-sig. Crucially, hinTS supports **weighted thresholds** natively, enabling expertise-based voting without overhead. (IACR) The extension Dyna-hinTS (ePrint 2025/631) supports dynamic committee rotation without re-setup.

Semantic Intent Commitment via Pedersen vector commitments solves the "committing to natural language" problem. No direct academic construction exists, but the building blocks compose cleanly: a user's prompt passes through a sentence embedding model (e.g., all-MiniLM-L6-v2) producing a 384-dimensional float vector, quantized to 16-bit fixed-point, then committed as $C = \text{Pedersen_vector_commit}(\text{quantized_embedding}, r)$ on BN254. A Groth16 proof then

demonstrates cosine similarity between the committed embedding and the AI output embedding exceeds threshold τ — requiring approximately **15K-30K R1CS constraints**, generating proofs in 3-5 seconds. This is the mechanism that cryptographically encodes the illocutionary fidelity gap.

Inner-Product Functional Encryption (Ernst & Mitrokotsa, ACNS 2023) enables selective disclosure of behavioral biometric properties. Using function-hiding IPFE, PragmaJudge can prove "typing speed variance is within human range" or "keystroke timing pattern matches enrolled user within threshold ϵ " without revealing raw timing data. The inner-product structure naturally maps to distance computations on behavioral feature vectors, (IACR) (ACM Digital Library) and many IPFE schemes operate over the same pairing-friendly groups as Groth16/BN254.

Federated learning creates a privacy-preserving biometric intelligence network

The most novel architectural pattern for PragmaJudge's biometric model training is **vertical federated learning across signal modalities** — where different biometric signals from different physical devices train a shared model without any single device holding all modalities.

BVFLMSP (Kar et al., April 2026) introduces a Bayesian vertical FL architecture where each client models a specific data modality with a Bayesian neural network, providing uncertainty estimates alongside predictions. (arXiv) For PragmaJudge, this maps directly: the keyboard captures keystroke dynamics (Client 1), the mouse captures cursor trajectories (Client 2), a GSR wearable captures electrodermal activity (Client 3), and the device IMU captures physiological tremor (Client 4). Each device runs a local encoder producing intermediate representations, which W3bstream prover nodes aggregate using split learning with differential privacy noise added to embeddings before transmission. (arXiv)

The recommended architecture uses a **split model design**: shared feature encoder layers (aggregated via FL) learn general human behavioral patterns, while personal classifier layers (staying on-device, never shared) learn user-specific biometric signatures. The FedFV framework (Mi-research) demonstrated this approach achieves **EER as low as 0.07%** on biometric authentication. Practical differential privacy guarantees at $\epsilon = 4$ introduce only **2-5% accuracy degradation**. (MDPI) A compute-node-assisted secure aggregation protocol (Xu et al., 2025) reduces aggregation overhead to **1.25% of baseline** — and W3bstream's prover nodes map directly to the compute node role.

IoTeX's existing Scaleout partnership validates this approach. Scaleout already develops protocols for contributor-owned ML models trained on verifiable Pebble Tracker data (Starter) with Initial Model Offerings (IMOs) enabling community co-ownership. (IoTeX Blog) PragmaJudge extends this from GPS/climate data to behavioral biometrics, creating the first **DePIN biometric intelligence network** where players collectively improve anti-cheat models while retaining ownership of their physiological data.

On-device inference is practical: keystroke and mouse dynamics models require feature vectors of only 43-100 dimensions. (MDPI) LSTM or 1D-CNN models quantized to INT8 fit in under **500KB**, running inference in under **500ms at less than 1mW power** — well within TinyML constraints for mobile and IoT devices. (PubMed Central)

Mechanism design for a 36-agent epistemic economy

The research reveals a critical design tension that shapes PragmaJudge's entire incentive architecture: **quadratic voting fails for information aggregation under uncertainty** (Benham, Hemenway Falk & Tsoukalas, Management Science, 2024). QV's known advantages in preference aggregation do not extend to common-value information aggregation because cost convexity disincentivizes better-informed voters from adequately conveying private information. (INFORMS) (SSRN) This means PragmaJudge must use QV for **value decisions** (which evaluation rubrics matter) but **not** for factual quality assessment.

For quality assessment, **Robust Bayesian Truth Serum (RBTS)** provides the strongest incentive-compatible mechanism. RBTS achieves strict Bayes-Nash incentive compatibility for any $n \geq 3$ (AAAI) — PragmaJudge's 36 agents far exceed this minimum. Each agent submits a quality evaluation score plus a prediction of other agents' scores. Agents are rewarded when their evaluation is "surprisingly common" (Peercommunityin) (BTS information score) and when their prediction is accurate (quadratic prediction score). (PubMed) The dominantly truthful multi-task variant (Kong, 2020-2024, Journal of the ACM) provides the strongest known truthfulness guarantee when agents evaluate multiple AI outputs per session.

The economic enforcement layer should implement **STAKESURE-style slashing** (arXiv 2401.05797), which redistributes slashed funds to harmed parties rather than burning them. (arXiv) Combined with **AgentBound Tokens** (Chaffer, 2025) — non-transferable tokens bound to AI agents analogous to soulbound tokens for humans — this creates a skin-in-the-game accountability chain. Each agent stakes IOTX tokens plus holds a soulbound ABT credential; slashing triggers when an agent's evaluation deviates significantly from validated consensus, (arXiv) with correlated slashing curves that penalize leniently for isolated disagreements but heavily for coordinated deviation.

A formal finding constrains the design: **no collusion-resistant mechanisms exist in general** under signal manipulation conditions (ResearchGate) (AAAI) (2024 DePIN mechanism design paper). Mitigation requires combining multiple overlapping mechanisms: random agent assignment per evaluation, commit-reveal schemes preventing pre-coordination, temporal isolation of evaluations, Kleros-style escalating appeals (panels of 7 → 15 → 36), and non-transferable reputation tracked via soulbound credentials.

Mobile biometrics close the Layer 3 anti-cheat gap

The mobile gaming market generates **\$92 billion annually** with **3.3 billion players**, yet **86% of mobile games have been compromised** (Irdeto) while only **22% of developers employ anti-cheat**. The current anti-cheat stack has two layers — hardware attestation (Apple App Attest, Android Play Integrity) and software integrity (ACE, Appdome) — but a critical **Layer 3 is entirely missing: continuous human physical presence verification**.

This gap is now existential. AI autonomous agents (Claude Computer Use, OpenAI Operator, Google Project Mariner) can tap, swipe, and interact with mobile apps like humans. (o-mega) They operate legitimate apps on real devices, **passing every existing anti-cheat check**. (arXiv) The autonomous AI agents market is projected to reach **\$44.7 billion by 2029**. A single operator can deploy hundreds of AI agents across cloud instances, each passing hardware attestation. This is the threat that only physiological biometrics can address.

The research identifies **BioMoTouch** (2025, arXiv) as the most promising mobile biometric framework: it fuses physiological features from capacitive-sensing signals with behavioral features from IMU measurements, achieving **EER of 0.27%** and **99.7% authentication accuracy**. (arXiv) Swipe gesture sequences of just 5 swipes achieve **EER of 0.2%**. (ScienceDirect) The AUToSen system (2024) achieves **F1-score of 98%** using accelerometer, gyroscope, and magnetometer for continuous authentication.

Hardware attestation creates the foundation layer. Apple's Secure Enclave generates non-exportable cryptographic keys with Apple-signed attestation certificates. (Security Boulevard +2) Android's StrongBox Keymaster uses a dedicated tamper-resistant security chip with Remote Key Provisioning (RKP) making leaked keyboxes technically impossible. (Tryigit) Both can serve as **hardware roots of trust for ioID registration**: generate an attested key in Secure Enclave/StrongBox → attest with Apple/Google servers → use the attested key to sign an ioID DID registration → verify attestation chain before creating the ioID NFT on IoTeX. This creates a cryptographic chain from silicon to blockchain.

Mobile devices actually offer **richer biometric signals than gaming controllers**: pressure-sensitive touchscreens, 6-axis IMU, proximity sensors, and capacitive contact patterns (Xpi) (ScienceDirect) versus controllers' more limited button-press and joystick inputs. The translation from VAPI's controller-based approach to mobile is architecturally natural — the same physiological tremor detection (8-12 Hz) works on smartphone accelerometers, (ACM Digital Library) (Nature) which use comparable hardware (Bosch BMI260/323 class) to controller IMUs. (IntechOpen)

Epistemic consensus beyond simple voting

The most architecturally significant consensus finding is **AgentAuditor** (arXiv:2602.09341,

2025), which replaces majority voting with structure-aware auditing. It constructs a "Reasoning Tree" of distinct hypotheses from agent evaluations, identifies Critical Divergence Points where agents disagree, and performs localized evidence audits at those points. On cases where the majority is wrong, AgentAuditor achieves **65.35% accuracy on GSM8K and 81.82% on AMC** — versus 0% for majority voting by construction. (arXiv) This is transformative for PragmaJudge: rather than counting votes, the system identifies *where* agents disagree and audits the reasoning at those specific points.

The 0.65 threshold consensus model formalizes cleanly in **graded epistemic logic**:

$T^{\{0.65\}}_{\{36\}}(\varphi)$ requires $\lceil 0.65 \times 36 \rceil = 24$ agents to believe φ . The simplicial models framework (Goubault et al., 2024) extends this to handle agent crashes using topological structures rather than Kripke models — directly relevant for a distributed fleet where agents may go offline.

The recommended consensus architecture layers five mechanisms:

- **Layer 0 (Byzantine foundation)**: Approximate agreement tolerating $f \leq 11$ Byzantine agents ($n=36 > 3 \times 11 + 1 = 34$), using graph-based convergence for ordinal evaluations (Dagstuhl)
- **Layer 1 (Liquid democracy)**: Per-domain delegation with a 15% weight concentration cap — an agent weak in literary analysis delegates to a specialist, but votes directly on factual accuracy
- **Layer 2 (Deliberation)**: CollabEval-style three-phase protocol — independent evaluation, structured panel debate (6 panels $\times$ 6 agents), cross-panel synthesis (Amazon) (arXiv) — with All-Agents Drafting to prevent early anchoring (arXiv)
- **Layer 3 (Weighted consensus)**: ReConcile-style confidence-calibrated voting (arXiv) where agents with better calibration (closer match between stated confidence and actual accuracy) receive weight bonuses
- **Layer 4 (Meta-governance)**: Futarchy-style prediction markets selecting which evaluation rubrics to apply, (ChainScore Labs) with agents staking reputation tokens on predicted outcomes (ChainScore Labs)

This stack mirrors IoTeX's own architecture: Roll-DPoS uses 36 delegate candidates (Messari) (Messari) with PBFT+VRF for block finality (Iotex) — PragmaJudge's 36-agent fleet is the epistemic analogue of IoTeX's consensus layer.

Speech acts and the legal case for cryptographic intent records

The philosophical framework that most precisely formalizes PragmaJudge's function is **speech act theory** — specifically the 2024 work of Williams and Bayne ("Chatting with Bots," *Inquiry*).

They introduce the concept of "**proto-assertion**": LLM chatbots occupy a liminal category between genuine asserters and mere information generators. (arXiv +2) Their outputs carry **quasi-illocutionary force** — they commit the system to something resembling truth-claims — but the absence of a clear "principal" behind these outputs creates an accountability vacuum.

(ResearchGate)

PragmaJudge fills this vacuum by formalizing each AI interaction as a three-layer cryptographic record aligned with Austin's tripartite distinction. **Layer 1 (Locutionary)**: (ScienceDirect) Poseidon hash of the literal prompt text. **Layer 2 (Illocutionary)**: Structured intent metadata classified by Searle's taxonomy (directive, assertive, commissive, expressive, declarative) (Discourseanalyzer) plus extracted Gricean implicatures. **Layer 3 (Perlocutionary)**: User-specified expected outcome criteria (Discourseanalyzer) committed via Pedersen vector commitment. The distance between Layer 2 and the AI's output constitutes the measurable illocutionary fidelity gap — PragmaJudge's core metric.

The Gricean implicature dimension is especially compelling because **LLMs systematically fail at it**. Only GPT-4 achieves human-level accuracy (94%) on implicature tasks; most models score 20-60%. (arXiv) A 2024 *Nature Scientific Reports* study found no evidence that LLMs have explainable representations of manner implicatures. (Nature) This creates direct market demand: PragmaJudge serves as external verification that an AI properly understood what was *implied*, not just what was literally stated.

The regulatory demand is concrete. **EU AI Act Article 12** mandates "automatic recording of events ('logs') while the high-risk AI systems is operating... to enable the tracing of the functioning of the AI system throughout its lifecycle." (EU Artificial Intelligence Act) (VeritasChain Blog)

This is *exactly* the cryptographic audit trail PragmaJudge provides. Article 13 requires transparency enabling deployers to "interpret a system's output and use it appropriately."

(EU Artificial Intelligence Act) Article 14 mandates documented human oversight mechanisms.

(European Commission) The EU Revised Product Liability Directive (2024/2853) now covers AI systems and requires lifecycle documentation. (Clifford Chance) In the US, despite federal deregulation under the Trump administration, **over 1,000 AI-related state bills** were introduced in 2024-2025, (Pillsbury Winthrop Shaw Pi...) and the Colorado AI Act creates a safe harbor for entities complying with NIST AI RMF (Privacy + Cyber + AI) — whose first trustworthy AI characteristic is "Accountable and Transparent." (NIST)

The *Moffatt v. Air Canada* (2024) ruling established that companies are liable for AI-generated information as if it were their own communication. (Just Security) PragmaJudge's cryptographic non-repudiation addresses the core evidentiary problem in this emerging liability landscape: **proving what was requested versus what was delivered**. No existing blockchain-AI system (VeritasChain, FLEXBLOK, ETHOS) includes a semantic intent verification layer — they handle data provenance and decision attribution but not illocutionary fidelity. (FlexBlok)

The unified PragmaJudge architecture

Weaving all eight domains together produces a coherent system where physical intelligence verification, cognitive AI accountability, and mobile DePIN infrastructure converge into a single protocol. The architecture operates in five integrated phases.

Phase 1 — Physical identity anchoring. A mobile device generates a cryptographic keypair in its Secure Enclave (iOS) or StrongBox (Android). Apple/Google attestation servers certify the key originates from genuine hardware. (adjoe +2) This attested key signs an ioID DID registration on IoTeX, creating a blockchain-anchored device identity. (Messari +5) The soulbound ERC-4671 credential binds the device identity to an enrolled behavioral biometric profile — keystroke timing distributions, (Wikipedia) swipe gesture characteristics, physiological tremor signatures, and GSR baseline — stored as Pedersen vector commitments, never as raw data. This establishes the hardware-rooted trust chain from silicon to blockchain that no AI agent can replicate.

Phase 2 — Intent commitment and cognitive monitoring. When a user submits a prompt to an AI system, PragmaJudge extracts structured intent using the speech act taxonomy (classifying the prompt as directive, assertive, etc.) and Gricean implicature analysis. The prompt's semantic embedding is committed on-chain via Pedersen vector commitment. Simultaneously, the user's device continuously collects behavioral biometric signals — keystroke dynamics, touch pressure, device IMU tremor, and (if available) GSR from a connected wearable. (IntechOpen) These signals flow through the vertical federated learning pipeline, where on-device TinyML models perform real-time cognitive state inference while contributing differential-privacy-protected gradient updates to the global model (ScienceDirect) via W3bstream secure aggregation. (arXiv)

Phase 3 — Multi-modal ground truth generation. As the AI delivers output and the user interacts with it, temporal cross-correlation across biometric modalities generates the cognitive ground truth signal. GSR onset timing relative to keystroke hesitation patterns and mouse trajectory changes produces a temporal binding signature that indicates genuine cognitive engagement versus passive acceptance. ZK-PoP Groth16 circuits verify that behavioral feature vectors fall within human population distributions and that the interaction pattern is consistent with deliberative evaluation — producing **192-byte proofs** verifiable in 8.2ms. Inner-product functional encryption enables selective disclosure: the proof demonstrates "this user's cognitive engagement level during evaluation was above threshold" without revealing any raw biometric data.

Phase 4 — Epistemic consensus and accountability verdict. The 36-agent fleet evaluates AI output fidelity through the layered consensus protocol. Agents first evaluate independently (All-Agents Drafting), then engage in structured panel debate (arXiv) (6 panels of 6), then cast expertise-weighted, confidence-calibrated votes. AgentAuditor-style reasoning trees identify critical divergence points for targeted analysis. (arXiv) RBTS scoring incentivizes truthful evaluation. The weighted 0.65 threshold (requiring ≥ 24 equivalent agent-votes) determines the

verdict. [hinTS](#) threshold BLS signatures aggregate ≥ 24 partial signatures into a single 896-byte SNARK-verified signature ([Hedera](#)) posted to IoTeX ([Hedera](#)) at constant 390K gas cost. The verdict includes the measured illocutionary fidelity gap — the cosine distance between the committed intent embedding and the AI output embedding, verified via Groth16 proof.

Phase 5 — On-chain settlement and reputation. The complete accountability record — intent commitment, behavioral biometric proof, cognitive engagement attestation, agent consensus verdict, and illocutionary fidelity score — settles on IoTeX L1 as a verifiable credential linked to the soulbound ERC-4671 token. STAKESURE-style slashing enforces agent accountability: agents whose evaluations consistently deviate from validated consensus lose staked IOTX, redistributed to harmed parties. ([arXiv](#)) Futarchy prediction markets govern rubric evolution. ([ChainScore Labs](#)) ([ChainScore Labs](#)) Token-curated registries maintain the qualified evaluator list. ([Mitosis University](#)) ([Bitcoin](#)) The accumulated record creates a portable, cross-platform "AI accountability passport" satisfying EU AI Act Article 12 logging mandates, ([EU Artificial Intelligence Act](#)) ([VeritasChain Blog](#)) NIST AI RMF transparency requirements, and emerging product liability documentation needs.

What this architecture makes possible that nothing else can

The integrated PragmaJudge architecture unlocks four capabilities that currently exist nowhere:

Involuntary cognitive accountability — using the human body's physiological response to AI output as unforgeable ground truth, combining GSR (arousal detection at 98.3% accuracy), ([ScienceDirect](#)) keystroke dynamics (cognitive engagement at 91% accuracy), and mouse trajectory analysis (decision quality inference at 68.8% variance explanation) ([PubMed Central](#)) in a temporal cross-correlation that no bot or AI agent can replicate.

Cryptographic intent fidelity — committing natural language intent as semantic embeddings via Pedersen vector commitments, then proving via Groth16 circuits that AI output satisfies the committed intent within a measured fidelity threshold — the first system to formally measure and cryptographically attest the gap between what was asked and what was delivered.

AI-agent-resistant human verification — addressing the existential threat that autonomous AI agents (Claude Computer Use, OpenAI Operator) pass all existing hardware attestation and software integrity checks ([Invezz +2](#)) by adding a physiological biometric layer that verifies continuous human physical presence at EER below 0.3%, anchored to hardware roots of trust via Secure Enclave/StrongBox → ioID chains.

Decentralized epistemic adjudication — moving beyond simple majority voting to a five-layer consensus stack (Byzantine approximate agreement ([Dagstuhl](#)) → liquid democracy delegation → structured deliberation ([Medium](#)) → confidence-calibrated weighted voting → futarchy meta-governance) that is formally analyzed to tolerate up to 11 Byzantine agents, incentivize truthful

evaluation via RBTS, and settle verdicts on-chain via 896-byte hinTS threshold signatures at constant gas cost.

PragmaJudge positions itself not as an anti-cheat system that happens to use blockchain, but as the **accountability layer for the physical-digital interface** — the protocol that proves a real human body, with measurable cognitive engagement, interacted with an AI system that delivered output within cryptographically committed intent boundaries. [CoinMarketCap](#) In a world where AI agents increasingly operate autonomously across mobile devices and digital platforms, [VentureBeat](#) this is not merely useful but foundational. The \$92 billion mobile gaming market provides the initial proving ground; the \$680 million → \$4.3 billion blockchain-AI market [SSRN](#) and the EU AI Act's compliance mandate [European Commission](#) provide the scale trajectory. IoTeX's DePIN thesis — that physical devices generating verifiable real-world data create economic value [IoTeX Blog +7](#) — finds its most sophisticated expression in PragmaJudge: [IoTeX Blog](#) [CoinMarketCap](#) the human body itself becomes the ultimate DePIN device, generating the one signal that no artificial system can produce.